



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to KAVIRAJ AGENCY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

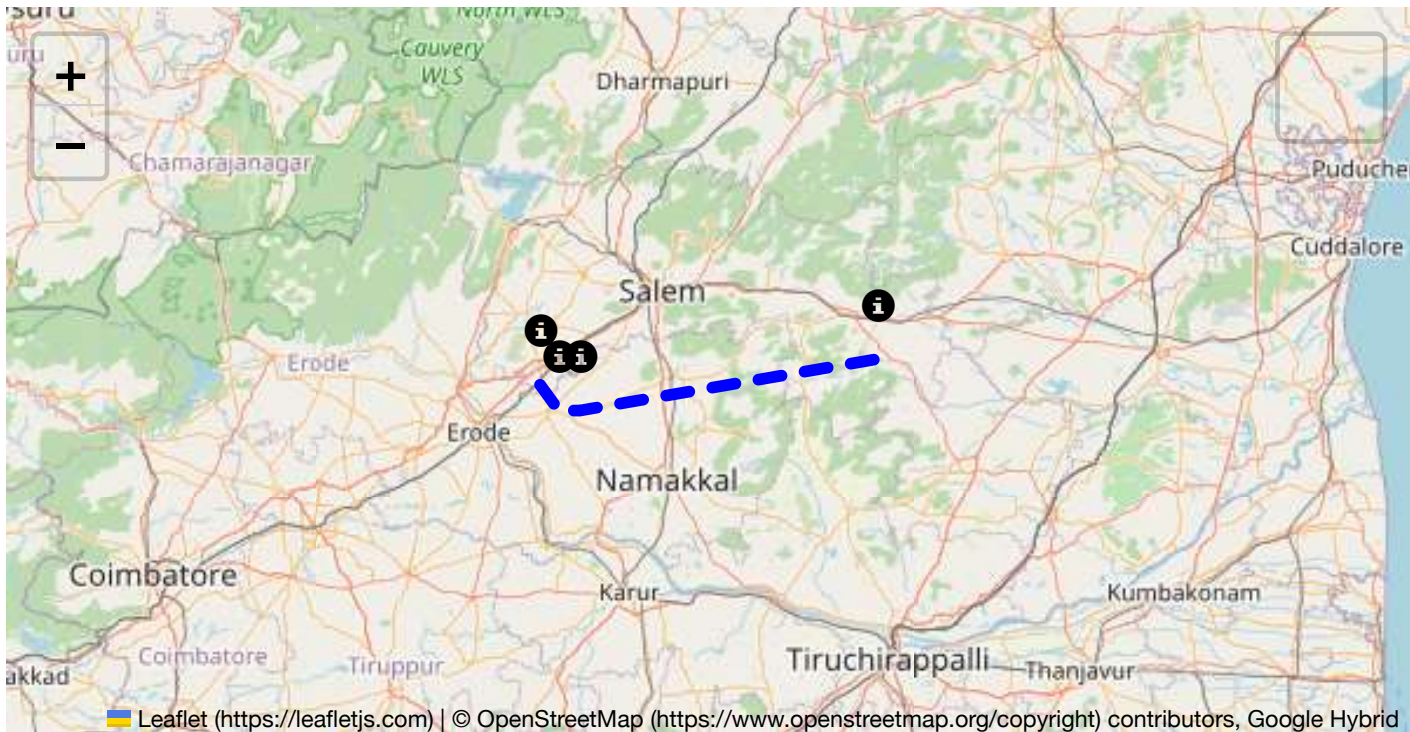
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 107.93 km
Estimated Duration: 2.6 hours
Adjusted Duration (Heavy Vehicle): 3.2 hours

Start: (11.4381, 77.8734)
End: (11.49824, 78.66449)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from CVQF+23W, Sangagiri, Tamil Nadu, to FMX7+9F2, Tamil Nadu, is approximately 107.93 kilometers. It includes major waypoints such as Namakkal and Kavundampalaiyam via State Highway 79 (SH 79). The route includes state highways and mostly rural roadways, passing through mixed-use areas comprising agricultural lands, small towns, and industrial zones.

2. Typical Weather Conditions and Hazards

Tamil Nadu experiences a tropical climate. The route may be affected by:

- **Monsoons (June to September):** Heavy rainfall can cause waterlogging and slippery roads.
- **Summer (March to May):** High temperatures can lead to road surface issues and tire blowouts.
- **Fog in Winter (December to February):** Reduced visibility, especially early in the morning.

3. Traffic Patterns

- **Peak Hours:** Morning (7 AM - 9 AM) and Evening (5 PM - 8 PM).
- **Congestion-Prone Areas:** Namakkal, being a busy junction, sees significant traffic. The SH 79 can be busy with commercial vehicles.

4. Assessment of Road Quality and Infrastructure

- **SH 79:** Generally well-maintained but watch out for occasional potholes and unmarked roadwork.
- **Rural Roads:** Can be narrow, with limited shoulder space.

5. Alternative Routes for Emergencies

- **Via NH948:** This offers an alternative if SH 79 is blocked. The national highway is generally wider and better maintained.
- **Local Roads Connectors:** Use local knowledge to skirt around specific incidents or roadblocks.

6. Local Regulations Affecting Hazardous Material Transport

- Permit requirements for hazardous goods.
- Restrictions during night travel.
- Limits on speed and designated rest areas for trucks.

7. Overview of Historical Incidents

- **Heavy vehicles:** Instances mostly involve overturning due to speed or road conditions, especially during rain.
- **Hazardous Materials:** Rare, with more focus needed on stringent checking and enforcement.

8. Environmental Considerations and Sensitive Areas

- Proximity to water bodies means potential spills can have severe consequences.
- Agricultural zones require care to prevent contamination of land and water.

9. Communication Coverage

- **Coverage:** Generally good along SH 79, but some rural stretches might have weak signals.
- **Dead Zones:** Identified in more isolated rural sections; advisable to have offline navigation tools.

10. Estimated Emergency Response Times

- **Urban Areas (e.g., Namakkal):** Approx. 15-30 minutes.
- **Rural Areas:** Response can be delayed up to an hour due to distance and road access issues.

11. Overall Summary of Risk Assessment

- **Low to Moderate Risk:** Good infrastructure on SH 79, but caution needed for weather conditions and rural segments.
- **Potential Hazards:** Weather-related, congestion at Namakkal, and poor road quality in parts.
- **Preparation:** Equip drivers with weather updates, proper training for handling emergencies, and backup communication devices.

This comprehensive analysis should help ensure safe transit for heavy vehicles transporting hazardous materials across these routes in Tamil Nadu.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44019, 77.87547	15 KM/Hr
3	Turn	Medium	11.39685, 77.88861	30 KM/Hr
4	Turn	Medium	11.38376, 77.89914	30 KM/Hr
5	Turn	High	11.38186, 77.90258	15 KM/Hr
6	Blind Spot	Blind Spot	11.37841, 77.90140	10 KM/Hr
7	Turn	Medium	11.36754, 77.93919	30 KM/Hr
8	Turn	Medium	11.41354, 78.08075	30 KM/Hr
9	Turn	High	11.41447, 78.08063	15 KM/Hr
10	Turn	High	11.41475, 78.08087	15 KM/Hr
11	Turn	High	11.42964, 78.11587	15 KM/Hr
12	Turn	Medium	11.42447, 78.15264	30 KM/Hr
13	Turn	Medium	11.43142, 78.18808	30 KM/Hr
14	Turn	Medium	11.43426, 78.20571	30 KM/Hr
15	Turn	Medium	11.43521, 78.20729	30 KM/Hr
16	Turn	High	11.45070, 78.20393	15 KM/Hr
17	Turn	High	11.45098, 78.20412	15 KM/Hr
18	Turn	Medium	11.44993, 78.22152	30 KM/Hr
19	Turn	High	11.44951, 78.22263	15 KM/Hr
20	Turn	High	11.44797, 78.22314	15 KM/Hr
21	Turn	Medium	11.44756, 78.22414	30 KM/Hr
22	Turn	Medium	11.45353, 78.23584	30 KM/Hr
23	Turn	Medium	11.49835, 78.32417	30 KM/Hr
24	Turn	Medium	11.50865, 78.33966	30 KM/Hr
25	Turn	High	11.51183, 78.34416	15 KM/Hr
26	Turn	Medium	11.51042, 78.34865	30 KM/Hr
27	Turn	Medium	11.50973, 78.34962	30 KM/Hr
28	Turn	Medium	11.50909, 78.35278	30 KM/Hr
29	Turn	Medium	11.48668, 78.36892	30 KM/Hr
30	Turn	Medium	11.47991, 78.38028	30 KM/Hr
31	Turn	Medium	11.46964, 78.39358	30 KM/Hr
32	Turn	Medium	11.46790, 78.39420	30 KM/Hr
33	Turn	High	11.46466, 78.39289	15 KM/Hr
34	Turn	Medium	11.46410, 78.39311	30 KM/Hr
35	Turn	Medium	11.45677, 78.40081	30 KM/Hr
36	Turn	High	11.45597, 78.40096	15 KM/Hr
37	Turn	Medium	11.45563, 78.40251	30 KM/Hr
38	Turn	Medium	11.45658, 78.40441	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
39	Turn	Medium	11.45458, 78.40886	30 KM/Hr
40	Turn	Medium	11.46740, 78.45082	30 KM/Hr
41	Turn	Medium	11.46728, 78.45442	30 KM/Hr
42	Turn	Medium	11.46798, 78.45523	30 KM/Hr
43	Turn	High	11.46798, 78.45844	15 KM/Hr
44	Blind Spot	Blind Spot	11.46857, 78.45844	10 KM/Hr
45	Turn	Medium	11.46835, 78.45959	30 KM/Hr
46	Turn	High	11.46800, 78.45987	15 KM/Hr
47	Turn	Medium	11.46461, 78.47397	30 KM/Hr
48	Turn	Medium	11.46419, 78.47445	30 KM/Hr
49	Turn	Medium	11.46211, 78.48159	30 KM/Hr
50	Turn	Medium	11.46184, 78.48270	30 KM/Hr
51	Turn	Medium	11.46106, 78.48357	30 KM/Hr
52	Blind Spot	Blind Spot	11.44233, 78.48592	10 KM/Hr
53	Turn	High	11.45641, 78.52757	15 KM/Hr
54	Turn	High	11.45723, 78.52761	15 KM/Hr
55	Turn	Medium	11.48770, 78.63405	30 KM/Hr
56	Turn	High	11.48870, 78.63452	15 KM/Hr
57	Turn	Medium	11.49431, 78.64705	30 KM/Hr
58	Turn	High	11.49697, 78.64738	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	Jayaa Hospital	11.3848223, 77.9021886	30 km/h	Medium
10	hospital	Government Hospital, Namagiripettai	11.4569484, 78.265263	30 km/h	Medium
11	hospital	Annai Hospital	11.4437321, 78.4858654	30 km/h	Medium
12	clinic	Shreetejo Ayurveda Clinic	11.444915, 78.485868	30 km/h	Medium
13	hospital	Vinola Hospital	11.4439899, 78.4857585	30 km/h	Medium
14	hospital	The Chirist Hospital	11.4443163, 78.4862134	30 km/h	Medium
15	clinic	Sabeetha Surgical Clinic	11.4421254, 78.4867012	30 km/h	Medium
16	hospital	Dr. Ramu Life Care Hospital	11.4426079, 78.4879762	30 km/h	Medium
17	hospital	Thai Hospital	11.4424776, 78.4894516	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
18	clinic	Jayaraj Memorial Clinic	11.4415635, 78.4890421	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
6	marketplace	Farmer Market	11.4146849, 78.0805463	30 km/h	Medium
7	school	Government Higher Secondary School	11.4150045, 78.0838172	30 km/h	Medium
8	school	Government school	11.4248027, 78.1323902	30 km/h	Medium
9	college	Paavai Arts And Science College For Women	11.4262443, 78.1546987	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44019, 77.87547



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.39685, 77.88861



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.38376, 77.89914



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.38186, 77.90258



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.37841, 77.90140



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.36754, 77.93919



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41354, 78.08075



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41447, 78.08063



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41475, 78.08087



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.42964, 78.11587



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.42447, 78.15264



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43142, 78.18808



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43521, 78.20729



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45070, 78.20393



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45098, 78.20412



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44993, 78.22152



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44951, 78.22263



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44797, 78.22314



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44756, 78.22414



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45353, 78.23584



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49835, 78.32417



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50865, 78.33966



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.51183, 78.34416



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.51042, 78.34865



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50973, 78.34962



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50909, 78.35278



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.47991, 78.38028



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46964, 78.39358



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46790, 78.39420



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46466, 78.39289



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46410, 78.39311



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45597, 78.40096



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45563, 78.40251



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45658, 78.40441



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45458, 78.40886



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46740, 78.45082



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46728, 78.45442



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46798, 78.45523



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46798, 78.45844



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.46857, 78.45844



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46835, 78.45959



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46800, 78.45987



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46461, 78.47397



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46419, 78.47445



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46211, 78.48159



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46184, 78.48270



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46106, 78.48357



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.44233, 78.48592



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45641, 78.52757



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45723, 78.52761



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.48770, 78.63405



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.48870, 78.63452



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49431, 78.64705



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49697, 78.64738
